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A mechanical model for McKibben pneumatic artificial muscles based on limiting chain extensibility and 3D application of the network alteration theories

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ABSTRACT

Pneumatic Artificial Muscles (PAMs) mimic the behavior of skeletal muscles by generating contractile force when they are pressurized. Stiffness of these actuators depends on the applied pressure because the inner elastomeric tube exhibits non-linear mechanical behavior which also makes control of the actuator difficult. Therefore, it is crucial to obtain a precise mechanical model for these actuators. In this work, based on the theory of limiting chain extensibility, a new continuum mechanics-based model is developed for elastomeric McKibben PAMs to predict stiffness and output parameters such as free contraction, blocked force, and dead-band pressure during the actuation course. The developed model is consistent with network alternation theories, which allow predicting the softening observed in first cycles of inflation-deflation (Mullins effect). The established relations can predict variations of the actuation force due to the alternation of the material network parameters as a result of Mullins softening. In order to determine the material parameters of the bladder, uniaxial tensile tests have been conducted on a virgin silicon rubber. Cyclic tests have also been conducted on the fabricated PAMs in virgin and completely softened states to obtain their characteristic curves. It is concluded that, fractional evolution laws can be well combined with the developed model to predict the behavior of PAMs during cyclic deformations. It is observed that, even for contractions less than 25%, the maximum principal stretch in a PAM can exceed three which implies that, simple strain energy functions such as NeoHookean and Mooney-Rivlin should not be employed for PAMs. It is also concluded that, the Mullins softening increases the free contraction while makes no remarkable effect on the blocked force.

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1. Introduction

Pneumatic Artificial Muscles (PAMs) mimic the behavior of skeletal muscles by generating contractile force when they are pressurized (Andrikopoulos et al., 2011). McKibben muscle is the most popular type of PAMs, which consists of an inner tube covered by a braided shell (Daerden and Lefeber, 2002). These actuators are difficult to control due to the highly non-linear behavior of the inner tube, compressibility of air and complicated geometry of the braid (Madden, 2004). The inner tube is usually made of an elastomeric material which undergoes large deformations under relatively small pressures. The threads of shell are almost inextensible, hence inflation of the tube results in shortening of the shell which produces pulling force if the muscle is coupled to a mechanical load, as illustrated in Fig. 1.

PAMs offer special advantages, including high power to weight/volume ratio, flexibility, size availability and low cost (Ganguly, 2012). These characteristics make them ideal for human interaction. Therefore, they are widely used as a source of actuation in devices designed for industrial, aerospace, medical, biorobotics, rehabilitation and mobility assistance applications (Asiaban, 2020; Fadaiepour et al., 2017). In (Lee and Shimoyama, 1999), a miniaturized PAM has been employed in an artificial hand with an acceptable soft movement profile. In (Wereley, 2009) a PAM with swaged end fittings is used in a morphing aircraft where the PAM is tested for 120 million loading cycles without failure.

Due to the non-linear mechanical behavior of elastomers at large deformations, the stiffness of these actuators would depend on the applied pressure. Therefore, it is crucial to obtain a precise mechanical model for them. Modeling of the mechanical behavior of these actuators is generally done by two different approaches. The first approach is based on empirical investigations regardless of the mechanical nature of the inner tube and type of the used ele-

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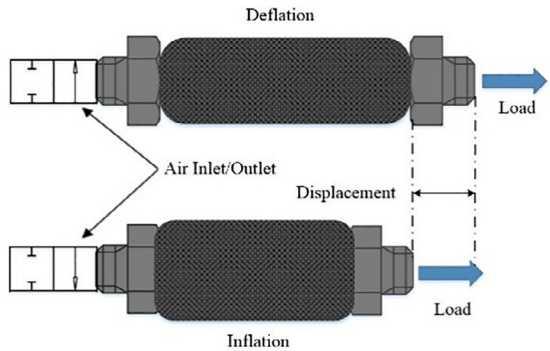

Fig. 1. A McKibben pneumatic artificial muscle.

ments. This approach tries to describe the relationship between applied pressure and produced force using properties of the PAM. The first force analysis of the double helix braided pneumatic muscles by Gaylord in 1958 (Gaylord and Fluid, , 1958) falls in this category. Wickramatunge and Leephakpreeda developed an empirical model to describe the mechanical behavior of PAMs based on characteristics of a mechanical spring calibrated using real experiments (Wickramatunge and Leephakpreeda, 2010). Song et al. developed a new Artificial Neural Network (ANN) to model the non-linear behavior of PAMs (Song, 2015). Reynolds proposed a three-element model for PAMs consisting of a contractile force-generating element, a spring element, and a damping element in parallel (Reynolds, 2003). Zhang et al. used the finite element method to simulate the non-linear dynamic behavior of PAMs (Zhang et al., 2005).

The second approach takes care of the mechanical nature of the inner tube and the surrounding element. Some models use the concept of virtual work and are based on the energy conservation principle (Klute and Hannaford, 2000). This approach involves more detailed force analysis. The model proposed by Chou and Hannaford is a typical example (Chou and Hannaford, 1996). Doumit et al. proposed a new static model which relates the actuation force to inflation pressure as well as geometric properties of the tube (Doumit et al., 2009). Wang et al. developed a quasi-static model to consider the elastic energy effect on PAM force. They applied force balance approach to derive the equilibrium equations (Wang et al., 2015).

In this work, based on the theory of limiting chain extensibility (LCE), a new continuum mechanics-based model is developed for elastomeric McKibben PAMs to predict their stiffness and output parameters such as free contraction, blocked force, and dead-band pressure during the actuation course. The developed model is consistent with network alternation theories (NAT), which allows predicting the softening occurred in the first cycles of inflation-deflation i.e., the Mullins effect. To adapt the proposed constitutive relations with NAT, the Gent strain energy function (SEF) is used which allows inserting the evolution of the material network in the constitutive relations though making the relations mathematically complicated. The established relations allow predicting variations of the outputs due to the alternation of the material parameters as a result of the Mullins softening. Moreover, some McKibben PAMs are fabricated using a silicon rubber as the bladder. Uniaxial tensile tests are conducted on standard virgin samples of the bladder in order to determine the model material parameters. Cyclic tests are also performed on the fabricated PAMs in both virgin and softened states to obtain the characteristic curves and compare the acquired test data with predictions of the proposed model.

2. Mechanical model

Elastomeric tube of a PAM can be assumed as a thick-walled circular cylinder, as shown in Fig. 2. The tube is described in the initial configuration by (Ogden, 1972):

$$A \leqslant R \leqslant B, \quad 0 \leqslant \Theta \leqslant 2\pi, \quad 0 \leqslant Z \leqslant L \tag{1}$$

where (R, Θ, Z) are the cylindrical coordinates in the initial (undeformed) configuration, A and B are the initial inner and outer radii, respectively, and L is the initial length of the tube.

Neglecting the end cap effects on the deformation and stress concentration (Khajehsaeid, 2016; Tsagarakis and Caldwell, 2000.; Khajehsaeid and Reese, 2015) we can assume that the cylinder remains a cylinder after inflation so the deformed configuration of the tube under internal pressure is described by:

$$a \leqslant r \leqslant b, \quad 0 \leqslant \theta \leqslant 2\pi, \quad 0 \leqslant z \leqslant l \tag{2}$$

where (r, θ, z) are the cylindrical coordinates in the deformed configuration, a and b are the current inner and outer radii, respectively, and l is the current length of the tube. Assuming axisymmetric deformation we have:

$$r = f(R), \quad \theta = \Theta, \quad z = \lambda Z \tag{3}$$

where $\lambda = \frac{l}{L}$ is the axial stretch of the tube. Hyperelastic material models are often utilized to formulate the mechanical behavior of elastomers in the absence of hysteresis (Khajehsaeid, 2018a,b, 2015); hence the stress is calculated based on the associated strain energy potential function W (Khajehsaeid and Ramezani, 2014; Bechir, 2006) which is often expressed in terms of the invariants of the left Cauchy-Green deformation tensor $\mathbf{B}(= \mathbf{F}^{\mathrm{T}}\mathbf{F})$. In cylindrical coordinates, the deformation gradient $\mathbf{F}$ for the inflated inner tube is given by (Khajehsaeid, 2019; Soleymani and Khajehsaeid, 2017):

$$\mathbf{F} = \begin{bmatrix} \frac{dr}{dR} & 0 & 0 \\ 0 & \frac{r}{R}\frac{d\theta}{d\Theta} & 0 \\ 0 & 0 & \frac{dz}{dZ} \end{bmatrix} = \begin{bmatrix} \frac{dr}{dR} & 0 & 0 \\ 0 & \frac{r}{R} & 0 \\ 0 & 0 & \lambda \end{bmatrix} \tag{4}$$

Thus, the invariants of $\mathbf{B}$ are:

$$\begin{aligned} I_1 &= tr\,\mathbf{B} = {\lambda_r}^2 + {\lambda_\theta}^2 + {\lambda_z}^2 = \left(\tfrac{dr}{dR}\right)^2 + \left(\tfrac{r}{R}\right)^2 + \lambda^2 \\ I_2 &= \tfrac{1}{2}\left({I_1}^2 - tr\mathbf{B}^2\right) = \left(\tfrac{rdr}{RdR}\right)^2 + \lambda^2\left(\tfrac{dr}{dR}\right)^2 + \tfrac{\lambda^2 r^2}{R^2} \\ I_3 &= \det\mathbf{B} = \left(\tfrac{dr}{dR}\,\tfrac{r}{R}\,\lambda\right)^2 \end{aligned} \tag{5}$$

where λ_r, λ_θ and λ_z are the principal stretches. In the case of incompressibility which is a common assumption for elastomers, we have $I_3 = 1$[29]:

$$\frac{dr}{dR}\frac{r}{R}\lambda = 1 \tag{6}$$

Using (6), the invariants can be rewritten:

$$\begin{aligned} I_1 &= \left(\tfrac{R}{\lambda r}\right)^2 + \left(\tfrac{r}{R}\right)^2 + \lambda^2 \\ I_2 &= \tfrac{1}{\lambda^2} + \tfrac{R^2}{r^2} + \tfrac{\lambda^2 r^2}{R^2} \\ I_3 &= 1 \end{aligned} \tag{7}$$

Integrating (6), the radial deformation $f(R)$ is obtained:

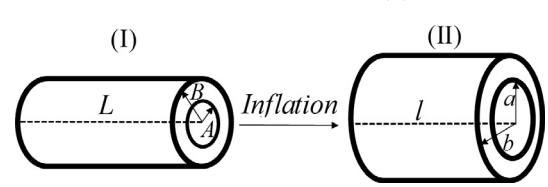

Fig. 2. The inner tube in (I) initial configuration, and (II) deformed configuration.

$$r = f(R) = \sqrt{\frac{R^2}{\lambda} + K} \tag{8}$$

where $K$is an integration constant which should be calculated regarding the boundary conditions. Using (6), $\mathbf{B}$can be rewritten as:

$$\mathbf{B} = \begin{bmatrix} \left(\frac{R}{\lambda r}\right)^2 & 0 & 0 \\ 0 & \left(\frac{r}{R}\right)^2 & 0 \\ 0 & 0 & \lambda^2 \end{bmatrix} \tag{9}$$

Due to the incompressible and isotropic behavior of elastomers, the Cauchy stress is given by [30,31]:

$$\boldsymbol{\sigma} = 2\mathbf{B}\frac{\partial W}{\partial \mathbf{B}} - P_s\mathbf{I} = 2\mathbf{B}\left(\frac{\partial W}{\partial I_1}\frac{\partial I_1}{\partial \mathbf{B}} + \frac{\partial W}{\partial I_2}\frac{\partial I_2}{\partial \mathbf{B}}\right) - P_s\mathbf{I} \tag{10}$$

where P_s is a hydrostatic pressure associated with the incompressibility constraint and $\mathbf{I}$ is the second-order identity tensor. Using and (9) and (10), the Cauchy stress is calculated as:

$$\begin{aligned} \boldsymbol{\sigma} = -P_s \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + 2\frac{\partial W}{\partial I_1} \begin{bmatrix} \left(\frac{R}{\lambda r}\right)^2 & 0 & 0 \\ 0 & \left(\frac{r}{R}\right)^2 & 0 \\ 0 & 0 & \lambda^2 \end{bmatrix} \\ + \left(2\frac{\partial W}{\partial I_2}\left(\left(\frac{R}{\lambda r}\right)^2 + \left(\frac{r}{R}\right)^2 + \lambda^2\right) \begin{bmatrix} \left(\frac{R}{\lambda r}\right)^2 & 0 & 0 \\ 0 & \left(\frac{r}{R}\right)^2 & 0 \\ 0 & 0 & \lambda^2 \end{bmatrix} \right. \\ \left. - \begin{bmatrix} \left(\frac{R}{\lambda r}\right)^4 & 0 & 0 \\ 0 & \left(\frac{r}{R}\right)^4 & 0 \\ 0 & 0 & \lambda^4 \end{bmatrix} \right) \end{aligned} \tag{11}$$

One should choose an appropriate SEF to determine the components of stress [32,33]. This usually depends on several factors such as the range of strains, available experimental data for calibration of the model, etc. It is convenient to choose a SEF whose derivatives are constant with respect to the invariants of $\mathbf{B}$. According to (11), such a SEF would result in relatively simple stress components which facilitate the resulting equilibrium equations, thus determining P_s and K would be straight forward. Noticing this point, many researchers choose Moony-Rivlin SEF which often gives a reasonable accuracy for strains up to 200% which is usually assumed to be enough for PAM applications:

$$W_{MR} = C_{10}(I_1 - 3) + C_{01}(I_2 - 3) \tag{12}$$

where C_{10} and C_{01} are the model parameters. Unfortunately, there is no clear physical description for these parameters which makes it impossible to connect them to alteration of the material molecular network. To be able to connect the constitutive relations of PAMs with NAT [34], it is crucial to choose a hyperelastic model whose material parameters are related to the characteristic properties of the material. Arruda-Boyce 8-chain [35] and Gent models are typical examples of these models [36]. In this study we use the Gent model for its simple mathematical formulation:

$$W_G = -\frac{\mu J_m}{2}\ln\left(1 - \frac{I_1 - 3}{J_m}\right) \quad where \quad I_1 < J_m + 3 \tag{13}$$

where μ is the material shear modulus and J_m is the limiting value for $I_1 - 3$ taking into account the limiting chain extensibility [36,37]. Based on the physical description of the introduced material parameters, this SEF allows inserting the evolution laws of the material molecular network in the constitutive relations. According to (11) and (13), the stress components take the following forms:

$$\sigma_r = -P_s + \frac{J_m\mu}{J_m - (I_1 - 3)}\left(\frac{R}{\lambda r}\right)^2 \tag{14}$$

$$\sigma_\theta = -P_s + \frac{J_m\mu}{J_m - (I_1 - 3)}\left(\frac{r}{R}\right)^2 \tag{15}$$

$$\sigma_z = -P_s + \frac{J_m\mu}{J_m - (I_1 - 3)}\lambda^2 \tag{16}$$

Eqs. (14) to (16) along with (8) include two unknown parameters K and P_s which need at least two boundary conditions to be determined.

2.1. Displacement boundary condition

As shown in Fig. 2-(I), the PAM is in its resting length at the ambient pressure. Fig. 2-(II) shows the PAM in the inflated state which has a shorter length. The threads are assumed inextensible due to their high stiffness compared with the inner tube. As shown in Fig. 3, assuming that the lengths of threads remain constant during inflation, the outer radius of the tube after inflation can be determined using the following geometric equation [17]:

$$L^2 + (2\pi nB)^2 = l^2 + (2\pi nb)^2 \tag{17}$$

In (17), n is the number of turns of a single thread about the tube diameter [17]:

$$n = \frac{L}{2\pi B\tan\gamma_1} \tag{18}$$

where γ_1 is the angle between threads and the radial direction at rest configuration. Using Eq. (17), the current outer radius can be calculated [27]:

$$b = \sqrt{\frac{L^2(1 - \lambda^2) + 4\pi^2 n^2 B^2}{4\pi^2 n^2}} \tag{19}$$

Using (8) the below equations can be derived and solved for the current inner radius [27]:

$$\begin{cases} K = b^2 - \frac{1}{\lambda}B^2 \\ K = a^2 - \frac{1}{\lambda}A^2 \end{cases} \rightarrow a = \sqrt{b^2 - \frac{1}{\lambda}(B^2 - A^2)} \tag{20}$$

2.2. Force boundary condition

The equilibrium equation in the radial direction is written as follows (in the absence of body forces):

$$\frac{d\sigma_r}{dr} - \frac{1}{r}(\sigma_\theta - \sigma_r) = 0 \tag{21}$$

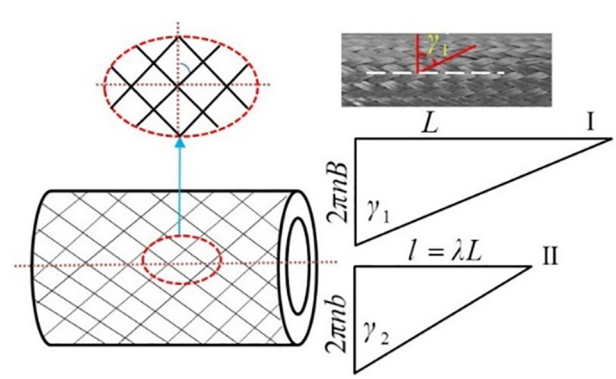

Fig. 3. General geometry of a PAM: (I) at the rest state, (II) during inflation.

Integrating (21), we have:

$$\int_a^r d\sigma_r = \sigma_r(r) - \sigma_r(a) = \int_a^r \frac{\sigma_\theta - \sigma_r}{r}\, dr \tag{22}$$

At the inner surface the radial stress is equal to the applied internal pressure:

$$\sigma_r(a) = -P \tag{23}$$

Right hand side of Eq. (22) can be rewritten using (14) and (15):

$$\int_a^r \frac{\sigma_\theta - \sigma_r}{r} dr = \int_a^r \left(\frac{J_m r \mu}{\lambda(r^2 - K)G} - \frac{J_m \mu (r^2 - K)}{\lambda r^3 G} \right) dr \tag{24}$$

where

$$G = \left(J_m - \frac{(r^2 - K)}{\lambda r^2} - \frac{r^2}{\lambda(r^2 - K)} - \lambda^2 + 3 \right) \tag{25}$$

Substituting (23) and (24) into (22) we have:

$$\sigma_r(r) = -P + \frac{\mu J_m}{4\Psi} \left(\ln\left(\frac{\Pi(r) + K\Psi}{\Pi(r) - K\Psi} \right) \Big|_a^r (J_m \lambda - \lambda^3 + 3\lambda + 2) + \Psi(4\ln(r) - \ln(\mathrm{E}(r)))|_a^r \right) \tag{26}$$

where

$$\Psi = \sqrt{\lambda^6 - 2J_m\lambda^4 + J_m^2\lambda^2 - 6\lambda^4 + 6\lambda^2 J_m + 9\lambda^2 - 4} \tag{27}$$

$$\Pi(r) = (2r^2 - K)\lambda^3 + (J_m + 3)(-2r^2 + K)\lambda + 4r^2 - 2K \tag{28}$$

$$\mathrm{E}(r) = \lambda^3 r^4 - J_m r^4 \lambda - K\lambda^3 r^2 - 3\lambda r^4 + 3K\lambda r^2 + 2r^4 - 2Kr^2 + K^2 \tag{29}$$

By subtracting (15) from (14) and using (26), the circumferential stress can be determined. The Circumferential force F_θ is also obtained by integrating the circumferential stress, as shown in Fig. 4.

$$F_\theta = l \int_a^b \sigma_\theta\, dr = l \int_a^b (\mathrm{S}_1 + \mathrm{S}_2)\, dr \tag{30}$$

$$\mathrm{S}_1 = \int_a^b \sigma_r\, dr \tag{31}$$

$$\mathrm{S}_2 = 2J_m \mu K \left(\frac{\tan^{-1}(\psi_N)}{\zeta_N} - \frac{\tan^{-1}(\psi_P)}{\zeta_P} \right) (\lambda^3 - 2J_m\lambda - 3\lambda + 2) \Big|_a^b \tag{32}$$

In (32), we have:

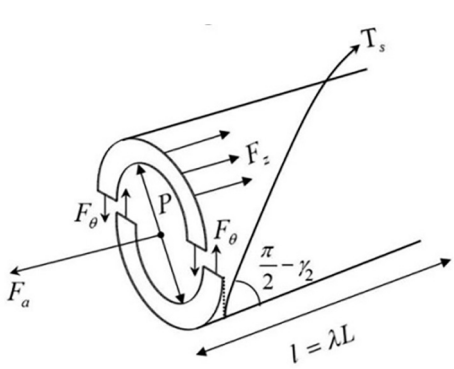

Fig. 4. Free body diagram of a pneumatic artificial muscle.

$$\psi_{P,N} = \frac{r\sqrt{2D}}{\sqrt{K\left(-D \pm \sqrt{D^2 - 4}\right)}} \tag{33}$$

$$\zeta_{P,N} = -D\sqrt{2K(D+4)\left(-D - 4 \pm \sqrt{(D+4)^2 - 4}\right)} \tag{34}$$

$$D = \lambda^3 - J_m\lambda - 3\lambda - 2 \tag{35}$$

Finally, the axial force is obtained by integrating (16):

$$F_z = \int_0^{2\pi} \int_a^b \sigma_z r\, dr d\theta = 2\pi \int_a^b (\mathrm{S}_1 + \mathrm{H}_2) dr \tag{36}$$

where

$$H_2 = \frac{J_m \mu r}{D+4} + \left(\frac{J_m \mu K}{(-D-4)^2} \right) - 6\lambda^3 - 6\mu\lambda + 2D - J_m^2\lambda^2 - 3D\lambda - J_m D\lambda - 3\lambda^6 + 4J_m\lambda^4 + 2J_m\lambda - 9\lambda^2 + 6\lambda + 12\lambda^4 \left(\frac{tan^{-1}(\psi_N)}{\zeta_N} - \frac{tan^{-1}(\psi_P)}{\zeta_P} \right) \Big|_a^b \tag{37}$$

Free body diagram of a typical PAM is shown in Fig. 4. The following equations are deduced by balancing the forces in the axial and circumferential directions [17]:

$$\begin{cases} \sum \hat{F}_\theta = 0 & \rightarrow \quad F_\theta + nT_s\cos\gamma_2 - Pal = 0 \\ \sum \hat{F}_z = 0 & \rightarrow F_z + T_s\sin\gamma_2 - F_a - P\pi a^2 = 0 \end{cases} \tag{38}$$

where γ_2 is the current angle between threads and the radial direction, T_s is the tension induced in the threads and F_a is the actuation force produced by the PAM. Solving the above equations for $F_a$we have:

$$F_a = F_z - P\pi a^2 + (Pal - F_\theta)\frac{\tan\gamma_2}{n} \tag{39}$$

Substituting (18) to (20), (30) and (36) in equation (39) the actuation force produced by the PAM is calculated as a function of the applied internal pressure (P), the current length (l) and the material parameters of the internal bladder. The developed relations in this section allow predicting variations of the actuation force and other outputs due to the alternation of the material parameters as a result of the material softening. Alternation of the material parameters due to the Mullins effect is studied in the context of the network alternation theories [38,39] which will be discussed in Section 4.

3. Results and discussion

In this section, the developed model is verified using experimental data acquired from testing the fabricated PAMs under different pressures. To determine the model parameters, uniaxial tensile tests (according to the standard ASTM-D412 type C) have been conducted on dumbbell-shaped samples of a virgin silicon rubber used as the inner tube of the PAMs. The results are shown in Fig. 5 in comparison with fittings by Moony-Rivlin and Gent models. Parameters of the models are reported in Table 1. Mooney-Rivlin (MR) model has been calibrated three times using the uniaxial tension data in ranges $\lambda \leqslant 1.5$, $\lambda \leqslant 2$ and $\lambda \leqslant 2.5$. It was observed that, each time the material parameters were totally different.

Equation (39) can be used to determine the required pressure for a desired contraction when the axial load is known. Fig. 6 shows the predicted pressure for 150 N axial load calculated using Gent and MR models in comparison with the experimental data acquired from the PAM shown in that figure with geometrical specifications reported in Table 1. It should be noted that, a

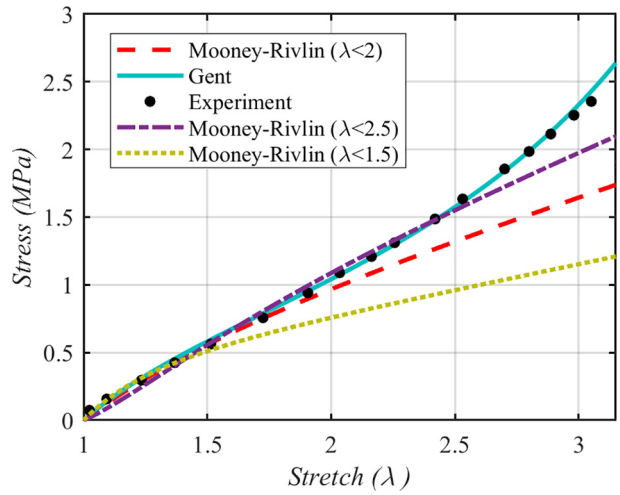

Fig. 5. Mooney–Rivlin and Gent models in comparison with the uniaxial test data.

Table 1
Properties and dimensions of the tested PAM.

Length at rest L (mm)	178.5
Outer radius of the tube B (mm)	4.8
Inner radius of the tube A (mm)	2.7
Braid angle at rest γ_1	69
Gent constant (virgin) μ (kPa)	535.4
Gent constant (softened) μ (kPa)	263.9
Gent constant (virgin) J_m	19.83
Gent constant (softened) J_m	34.33

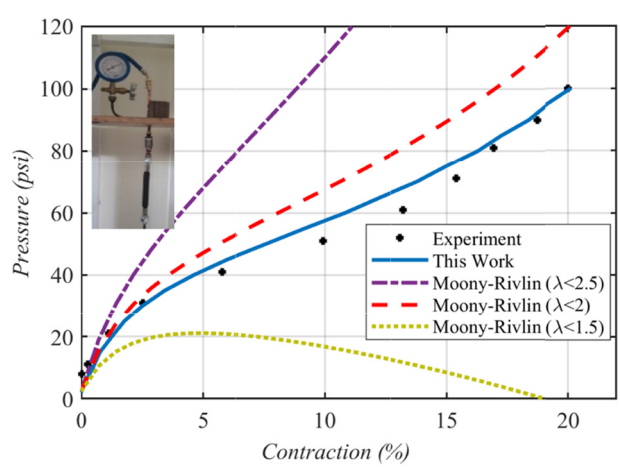

Fig. 6. Pressure-Contraction test on the virgin PAM under 150 N axial force: results of the proposed model (based on Gent SEF) in comparison with the model based on MR SEF [10] and the experimental data.

fixed-load method is utilized here to get the experimental data where the PAM is subjected to a fixed axial load to measure the axial contraction of the PAM at different applied pressure levels. It is evident that, results of the MR model significantly depend on the used stretch range for calibrating the model while it is not the case for the Gent model [35,40]. Since there is a 3D strain state in a pressurized PAM, the magnitude of strains is not pre-known. This makes the use of MR model problematic (despite its simple mathematical form) as one may not know which stretch range should be used for the model calibration. Consequently, the forthcoming results are calculated using the proposed model which is based on the Gent SEF. As seen in Fig. 6, a dead-pressure is also observed in the experiments which has been well predicted by the proposed model.

Fig. 7, shows the radial stress at the outer surface in different pressures as a function of the contraction ratio. As seen, $\sigma_r(r=b)$ is equal to internal pressure at zero contraction $(\lambda=1)$ which is a result of incompressibility. As the contraction ratio increases, $\sigma_r(r=b)$ changes from compressive regime to tensile regime. This regime change occurs at different contraction ratios as the internal pressure changes. Fig. 8 shows this critical contraction for different internal pressuers. Analogously, this behavior occurs in both circumferential and axial stresses as shown in Fig. 9. Circumferential and axial forces at the outer radius are shown in Fig. 10.

We can also plot the tube wall thickness with respect to the contraction at different internal pressures. Fig. 11, shows the wall thickness for 552 kPa internal pressure. The maximum wall thickness reduction occurs in the no-load state. This is an important point which should be taken into account in the design of PAM actuators.

Two main output parameters for a PAM are blocked force and free contraction. Given a certain pressure, these parameters point to the marginal states of the force-contraction curve. The angle of threads is an essential geometrical parameter which has a notable effect on these parameters. Fig. 12 shows the characteristic curve of the considered PAM for different thread angles. It is

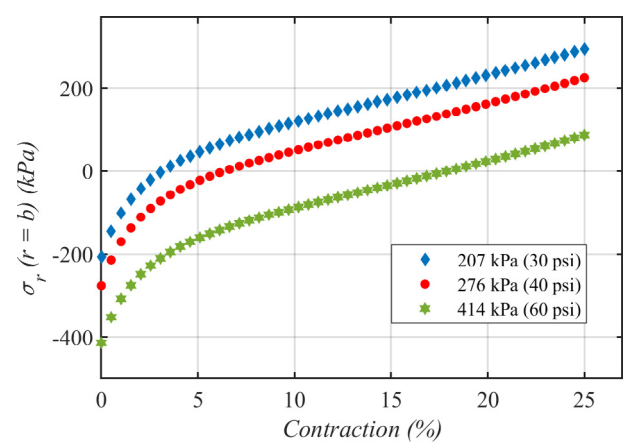

Fig. 7. Radial stress-contraction curve at the outer surface as a function of the inflation pressure.

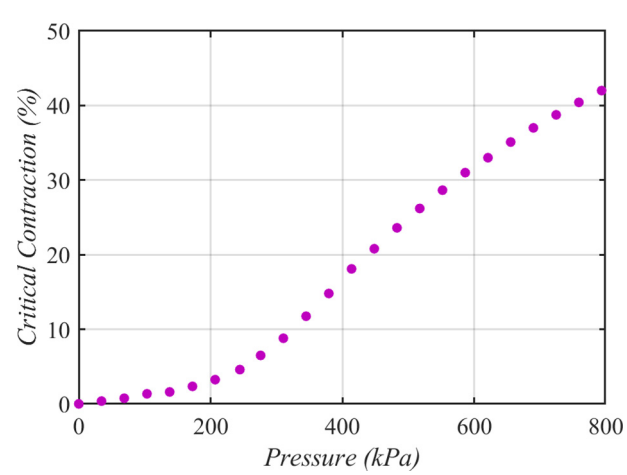

Fig. 8. Critical contraction for different inflation pressures.

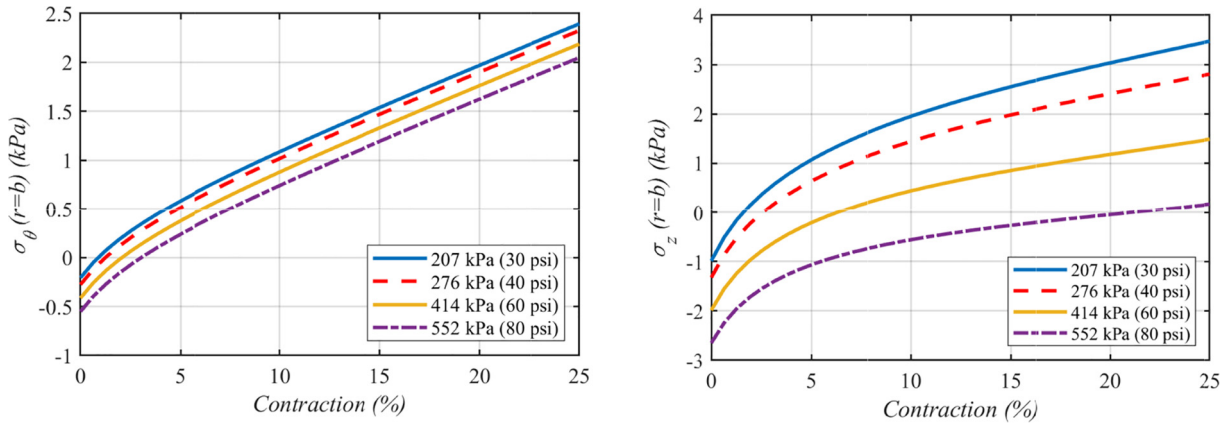

Fig. 9. Circumferential (left) and axial (right) stresses at the outer surface as a function of the inflation pressure.

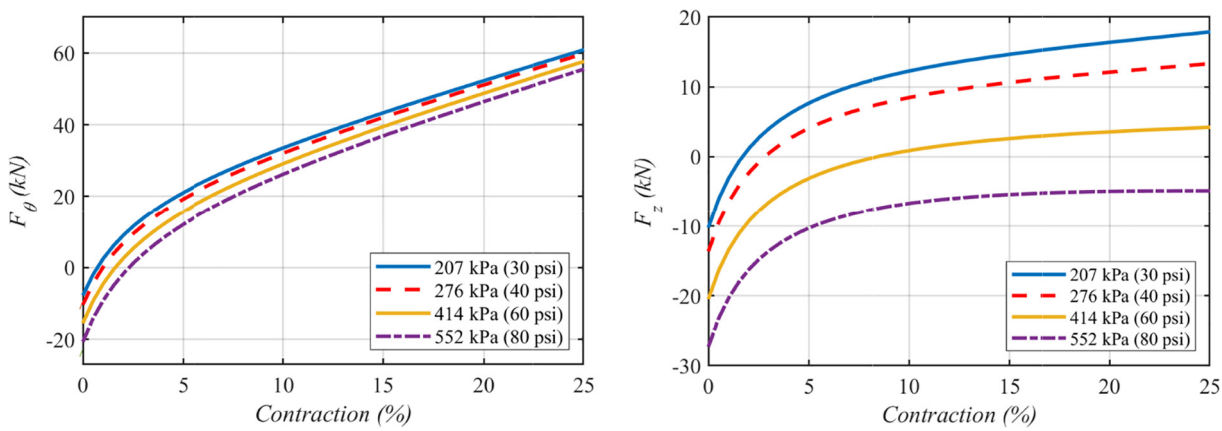

Fig. 10. Circumferential (left) and axial (right) forces as a function of the inflation pressure.

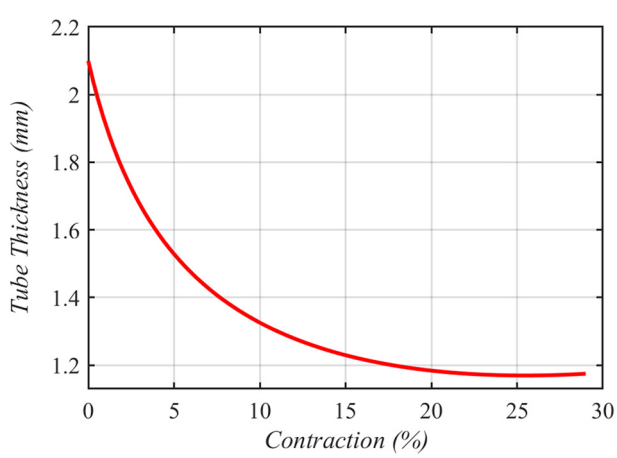

Fig. 11. Variation of the tube wall thickness for inflation pressure 552 kPa.

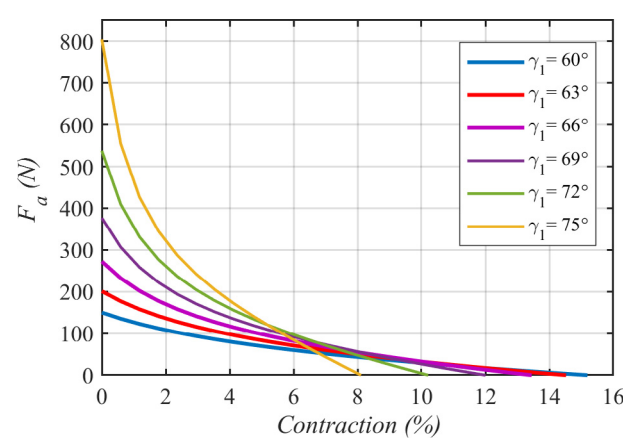

Fig. 12. Effect of the braid angel on blocked force and free contraction (P = 414 kPa).

observed that, increasing γ_1 affects the blocked force directly and the free contraction inversely.

The material parameters of the bladder also affect these outputs. Figs. 13 and 14 show the effects of μ and J_m on the actuation force at constant pressure (414 kPa). It is observed that, increasing μ (decreasing J_m) leads to fall of the free contraction. In other words, as high (low) as the shear modulus (chain extensibility limit), the lower the actuator contraction. Surprisingly, the blocked force is not much affected by alteration of the material parameters.

4. Mullins effect

Elastomers exhibit a softening in first cycles of loading called Mullins effect. This effect has been widely studied in the literature but there is still no unanimous interpretation for its physical nat-

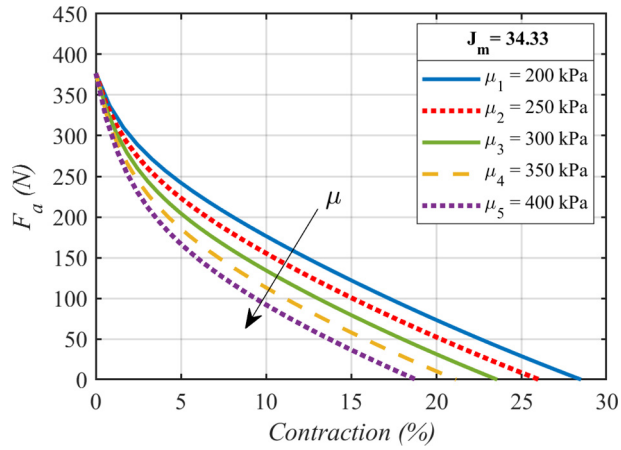

Fig. 13. Effect of μ on blocked force and free contraction (P = 414 kPa).

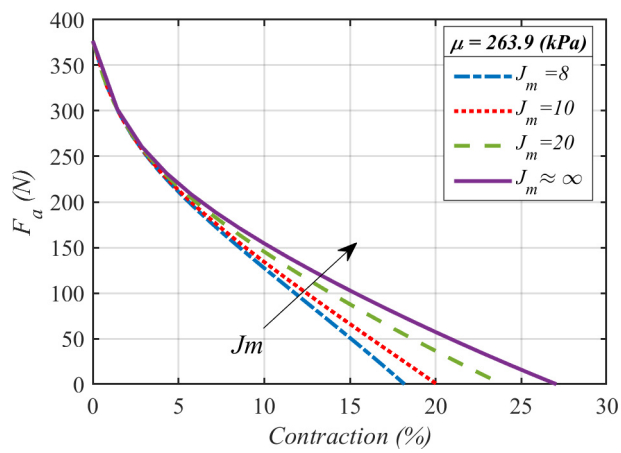

Fig. 14. Effect of J_m on blocked force and free contraction (P = 414 kPa).

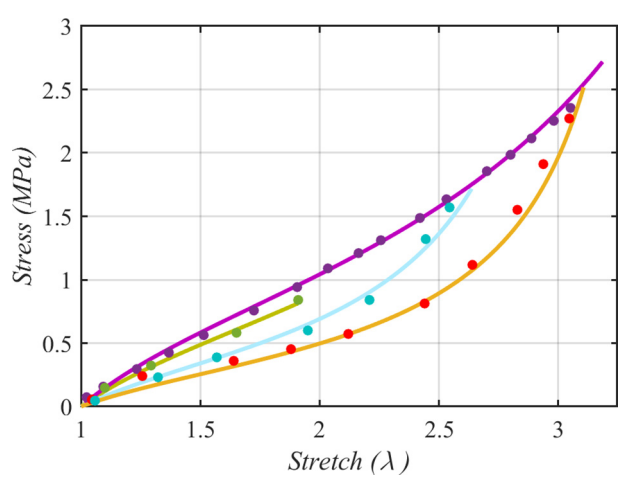

Fig. 15. Cyclic uniaxial tests on the core silicone rubber.

ure [41–44]. The network alteration theory which is employed in this work is one of the physical frameworks for explaining this effect.

Fig. 15 shows the cyclic uniaxial tests on a dumbbell-shaped sample of the utilized silicon rubber. Each curve has also been reproduced using the Gent model and the obtained material parameters μ and J_m are shown in Figs. 16 and 17 with respect to the maximum deformation ever experienced by the sample. Following the method conducted in [45], $I_1 - 3$, W, $\lambda_{\max} - 1$ and $\sqrt{\mathbf{B}.\mathbf{B}} - \sqrt{3}$ have been used as deformation measures here. In [45], based on the Langmuir theory, fractional laws have been proposed for evolution of the material network parameters:

$$\mu = \frac{1 + C_1\Phi}{C_2 + C_3\Phi} \tag{40}$$

$$J_m = \frac{1 + C_4\Phi}{C_5 + C_6\Phi} \tag{41}$$

where C_i are constant values and Φ is a threshold measure for softening. By means of the above relations, evolution of the material parameters are reproduced using different deformation measures. The results are reported in Figs. 16 and 17 where the obtained evolution laws are reported in Table 2. The limiting values of the material parameters in the undeformed and extremely deformed states are also calculated in Table 2. According to the obtained results $I_1 - 3$ and Wgive more reasonable results in the limiting states. The obtained laws will be later employed to predict the softening of the PAM during cyclic inflation-deflation tests. As shown in Figs. 16 and 17, the maximum principal stretch ($\lambda_{\max}$)exceeds 3 which implies that the MR SEF should not be employed when high pressures are applied to PAMs.

Cyclic tests have also been conducted on the fabricated PAMs. A virgin PAM was monotonically inflated up to 100 psi pressure under the application of fixed 150 N axial load. The axial contraction of the PAM was measured as the applied pressure increases. Another similar PAM was inflated in a step-wise manner with inflation history shown in Fig. 18 under the same axial load. The acquired experimental data are shown in Fig. 19.

The obtained evolution laws (reported in Table 2) can now be used along with the developed model i.e., Eqs. (39)–(41) to predict the pressure-contraction curves of the PAM under the inflation history shown in Fig. 18. The model predictions for the given inflation history is shown in Fig. 19 in comparison with the cyclic inflation-deflation tests data. It is observed that, the fractional laws well predict the evolution of material parameters during deformation provided that an appropriate threshold measure i.e., W is used in conjunction with these laws. Moreover, the obtained laws can be combined with the developed mechanical model for PAMs (i.e. Eq. (39)) to predict their behavior under cyclic deformations.

Fig. 20 compares the data from third cycle of inflation with predictions of the proposed model and a model based on the MR SEF from [10]. In both cases, Whas been used as the threshold measure. It is observed that, the developed model which is based on LCE is more accurate than models based on SEFs with constant derivatives with respect to I_1 and I_2 (e.g., NeoHookean and MR SEFs). As stated in the previous section, MR SEF is only accurate for $\lambda \leqslant 2$which is usually assumed to be sufficient for PAMs but we showed that, even for contractions less than 25% the maximum principal stretch in a PAM can exceed 3 which explains the inaccurate results of the models based on MR.

Figs. 21 and 22 show the characteristic curves of the tested PAM at different applied pressures before and after Mullins softening, respectively. It is observed that, the Mullins softening increases the free contraction while making no remarkable effect on the blocked force.

Fig. 23 compares the characteristic curves of the tested PAMs at 207 kPa (30 psi) and 414 kPa (60 psi) internal pressures for virgin and completely softened states. As shown, the free contraction is remarkably altered during the first cycles of loading where the blocked force is approximately unchanged.

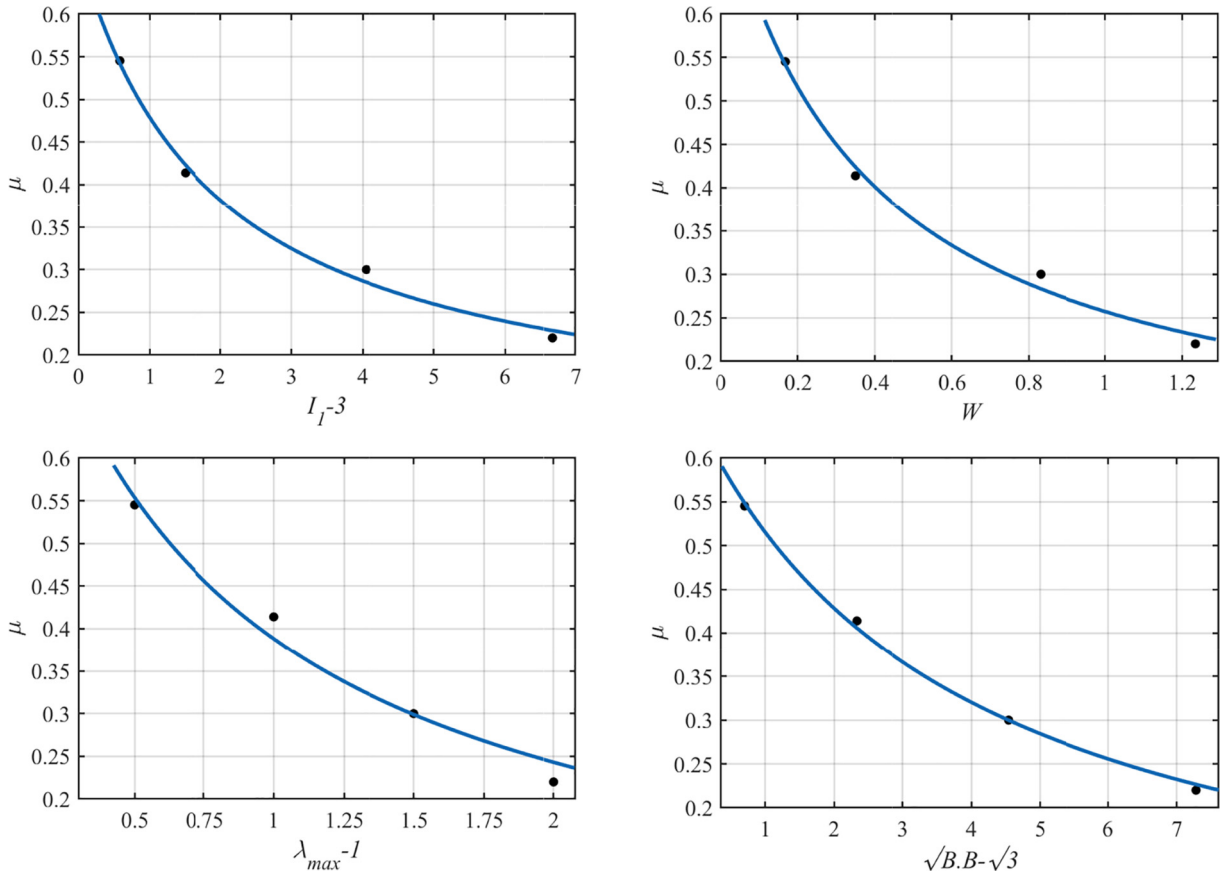

Fig. 16. Evolution of μduring deformation: experimental data vs. obtained evolution laws.

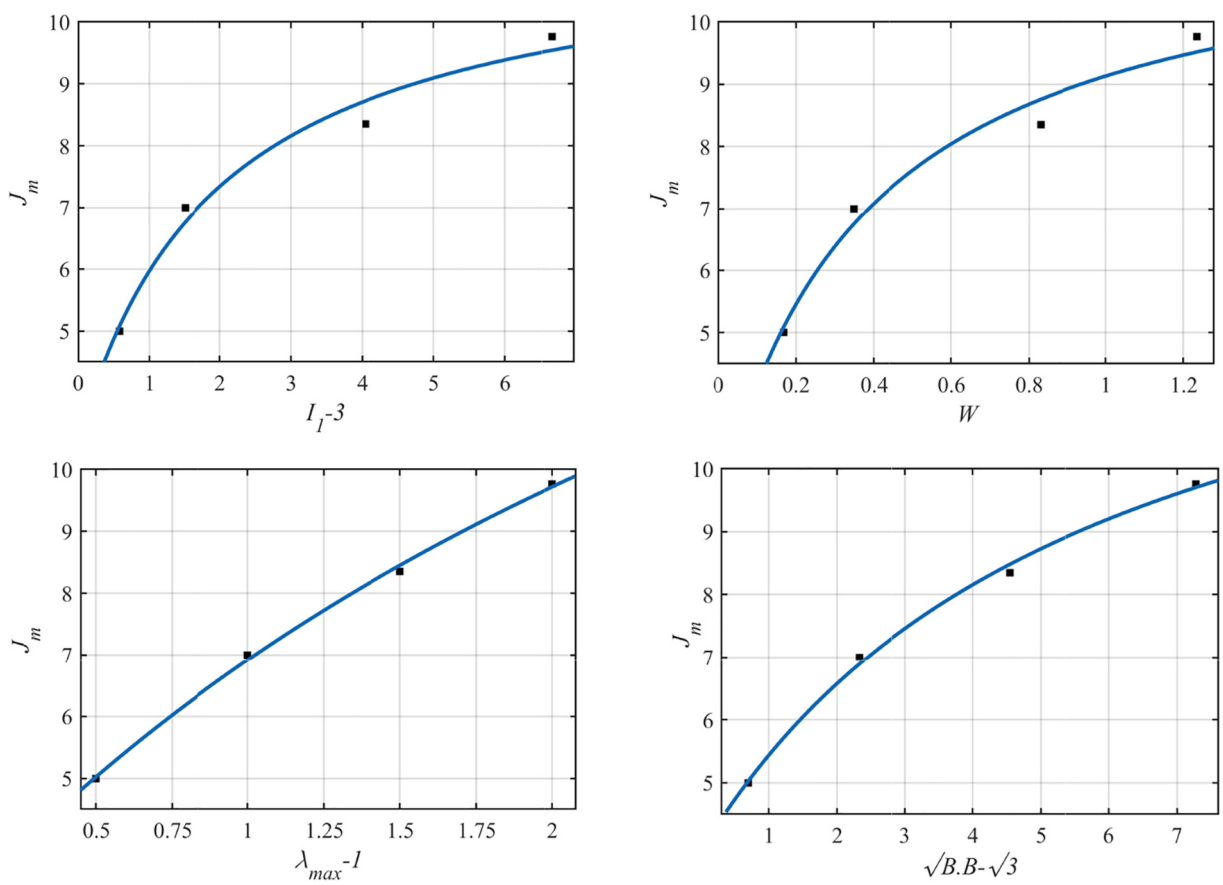

Fig. 17. Evolution of J_m during deformation: experimental data vs. obtained evolution laws.

Table 2
The evolution laws obtained for each deformation measure.

Φ	μ (*Mpa*)	$\lim\mu$ (*Mpa*)	J_m	$\lim J_m$
$I_1 - 3$	$\frac{1+0.077\Phi}{1.48+0.77\Phi}$	$\Phi \to 0 \;:\; \mu \to 0.675$ $\Phi \to \infty : \mu \to 0.099$	$\frac{1+1.79\Phi}{0.31+0.16\Phi}$	$\Phi \to 0 \;:\; J_m \to 3.221$ $\Phi \to \infty : J_m \to 11.18$
W	$\frac{1+0.251\Phi}{1.33+3.53\Phi}$	$\Phi \to 0 \;:\; \mu \to 0.752$ $\Phi \to \infty : \mu \to 0.071$	$\frac{1+14.6\Phi}{0.47+1.24\Phi}$	$\Phi \to 0 \;:\; J_m \to 2.127$ $\Phi \to \infty : J_m \to 11.77$
$\lambda_{\max} - 1$	$\frac{1+2.7e(-7)\Phi}{1.03+1.54\Phi}$	$\Phi \to 0 \;:\; \mu \to 0.97$ $\Phi \to \infty : \mu \to 0$	$\frac{1+2.33\Phi}{0.38+0.1\Phi}$	$\Phi \to 0 \;:\; J_m \to 2.631$ $\Phi \to \infty : J_m \to 23.31$
$\sqrt{\mathbf{B}.\mathbf{B}} - \sqrt{3}$	$\frac{1+5.2e(-9)\Phi}{1.55+0.4\Phi}$	$\Phi \to 0 \;:\; \mu \to 0.646$ $\Phi \to \infty : \mu \to 0$	$\frac{1+0.63\Phi}{0.26+0.044\Phi}$	$\Phi \to 0 \;:\; J_m \to 3.851$ $\Phi \to \infty : J_m \to 14.31$

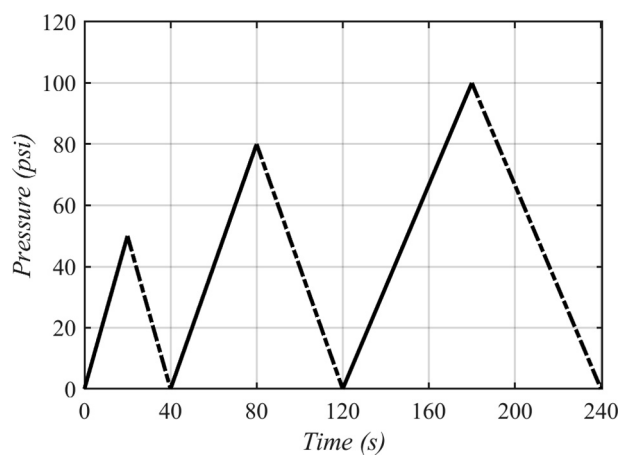

Fig. 18. The inflation history applied in cyclic inflation of the PAM.

5. Summary and conclusions

Based on the theory of limiting chain extensibility, a new continuum mechanics-based model was developed for elastomeric McKibben PAMs to predict their stiffness and output parameters during the actuation course. The developed model is consistent with network alternation theories which allow predicting the Mullins softening occurred in the first inflation-deflation cycles. The established relations were able to predict variations of the actuation force and other output parameters due to the alternation of the material parameters as a result of Mullins softening. Some PAM samples were fabricated using a silicon rubber as the internal bladder. Uniaxial tensile tests were conducted on the virgin samples of the silicon rubber to determine material parameters of the bladder. Cyclic tests were also conducted on the fabricated PAMs in virgin and also completely softened states. It is observed that, the fractional laws well predict the evolution of material parameters provided that an appropriate threshold measure is used in conjunction with them. These laws were then combined

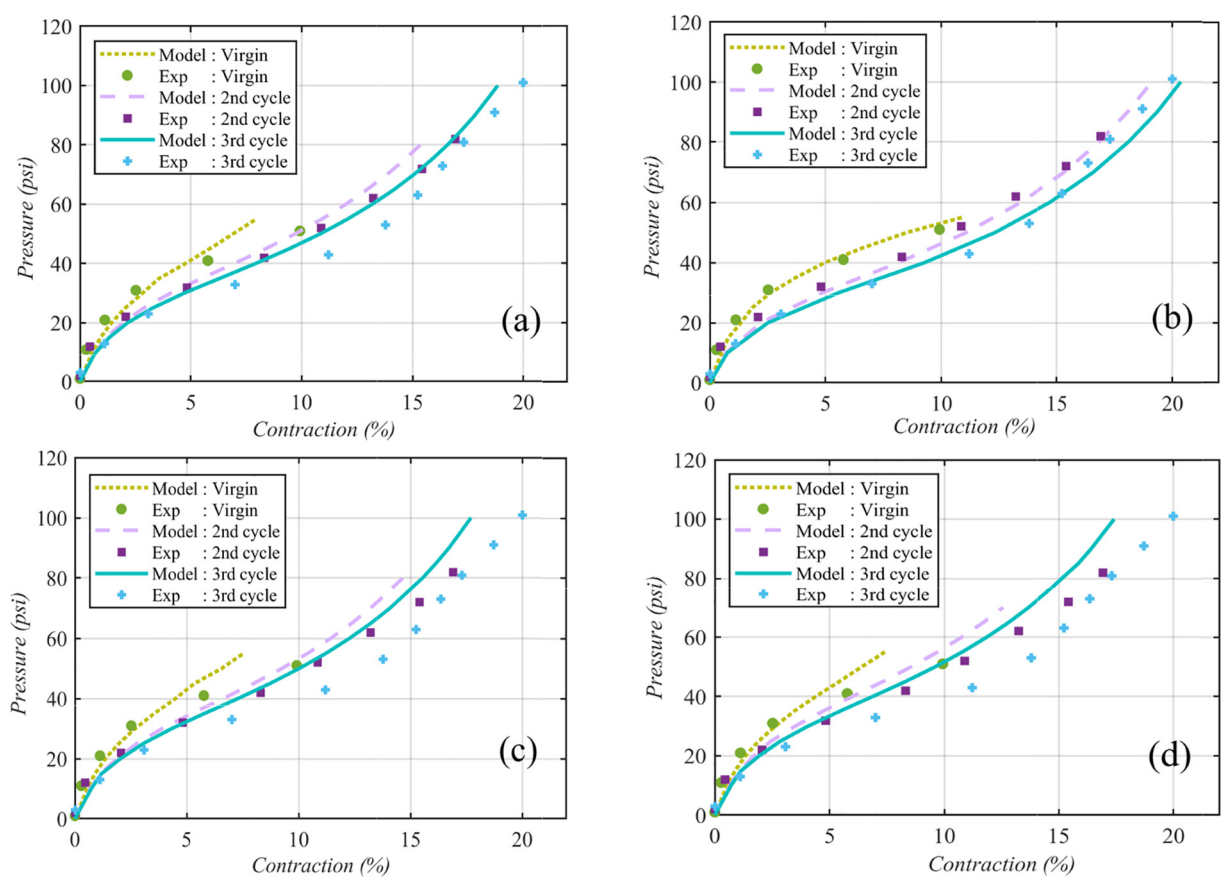

Fig. 19. Comparison between experiments and the model results obtained by means of different thresholds: a)$I_1 - 3$, b)W, c)$\lambda_{\max} - 1$and d)$\sqrt{\mathbf{B}.\mathbf{B}} - \sqrt{3}$.

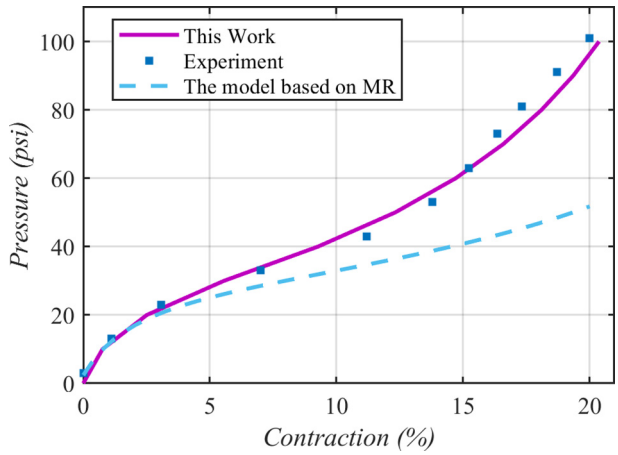

Fig. 20. Comparison of the data from third cycle of inflation with predictions of the proposed model (based on Gent SEF) and the model based on MR SEF [10].

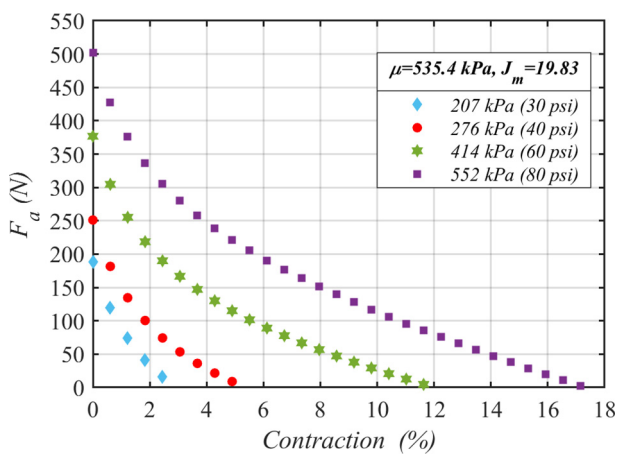

Fig. 21. Actuation force as a function of the inflation pressure for the virgin PAM.

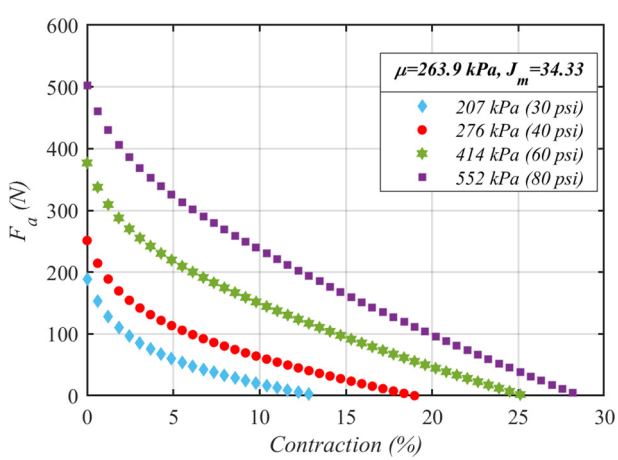

Fig. 22. Actuation force as a function of the inflation pressure for softened PAM.

with the developed model to predict the behavior of PAMs during cyclic deformations.

It was observed that, even for contractions less than 25%, the maximum principal stretch in a PAM could exceed 3 which implies that, simple SEFs such as NeoHookean and MR should not be

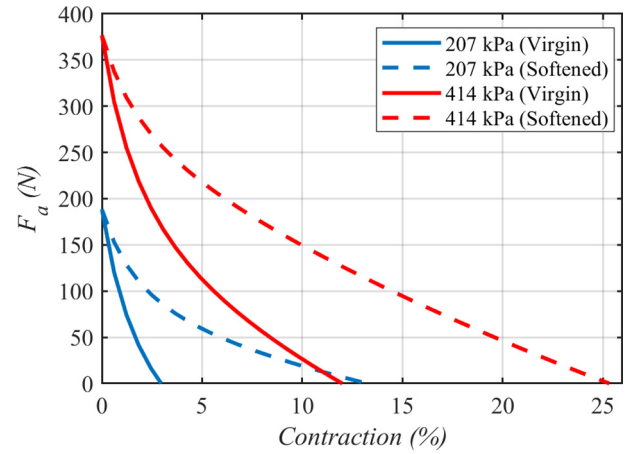

Fig. 23. Comparison between characteristic curves of the fabricated PAM at 207 kPa (30 psi) and 414 kPa (60 psi) internal pressures for virgin and completely softened states.

employed for PAMs. However, the developed model which is based on the limiting chain extensibility was shown to be more accurate than models based on SEFs with constant derivatives with respect to I_1 and I_2. It was also concluded that, the Mullins softening increases the free contraction while makes no remarkable effect on the blocked force.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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